

User Guide of DfW-prep

Byunguk Kim

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This document describes the purpose, required inputs, processing workflow, outputs, and key mathematical methods implemented in the DfW-prep pipeline. The pipeline converts raw UAV videos into a temporally resampled, georeferenced, and co-registered grayscale video product suitable for subsequent wave kinematics or bathymetry inversion analyses.

1. Program purpose and scope

DfW-Prep is a pre-processing pipeline that standardizes UAV imagery into a stable world-coordinate addresses two dominant sources of variability in field videos by explicitly separating attitude-induced distortion correction and platform-motion-induced image-plane transformation into modular processing stages. The pipeline combines attitude-only rectification, interactive ground control point (GCP) tracking, frame-wise georeferencing, and global co-registration onto a common grid.

2. Workflow

The pipeline runs the following stages in sequence:

- **Load:** read configuration and inputs; resample video to a target frame rate and cache it to disk.
- **Orthorectification:** apply a planar homography derived from pitch and roll to stabilize the horizon and reduce perspective skew.
- **GCP tracking:** select GCPs once, then track them through time with pyramidal Lucas-Kanade optical flow and anomaly detection.
- **Georeferencing & Warping:** estimate a 2D similarity (scaled rotation + translation) mapping from pixel GCPs to fixed world GCPs for each frame; optionally smooth GCP trajectories. Then, warp each frame to world-coordinates with a user-defined ground sampling distance (GSD), and record the per-frame world extent.
- **Visualization:** render a diagnostic movie that overlays warped frames on an optional basemap and GCPs.
- **Co-registration:** resample all warped frames onto a single global world grid, and

export a final standardized video and a global extent file, formatted for direct use as inputs to the DfW algorithm.

A compact flow representation is given below:

Raw videos (.mp4/.mov)

- temporal resampling to FPS and grayscale
- pitch-roll homography rectification
- GCP selection and optical-flow tracking
- frame-wise similarity transform estimation
- per-frame world warp and extent computation
- global-grid co-registration and export

3. Directory structure and required inputs

Each processing case must conform to a predefined directory structure to ensure consistent data handling. For a given case identified by the name *Case*, all input data are expected to reside in *Cases/Case/inputs*, while all generated outputs are written to *Cases/Case/outputs*.

Required input files (*Cases/Case/inputs*)

- **params.yaml**: A configuration file specifying the processing parameters. This includes, but is not limited to, the target frame rate (FPS), camera pitch and roll angles (PITCH, ROLL), focal length (FOCAL_LENGTH), desired ground sampling distance (GSD), and parameters controlling GCP tracking and temporal smoothing.
- **gcps.csv**: A comma-separated file containing the world-coordinate information of the ground control points. The pipeline interprets the numeric values in columns 2 and 3 as the planar world coordinates (Easting, Northing) associated with each GCP.
- **Video files**: One or more raw video files in .mp4 or .mov format, which constitute the primary observational input to the pipeline.

Optional input files (*Cases/Case/inputs*)

- **basemap.png**: A georeferenced or visually aligned basemap image, used exclusively for diagnostic and qualitative visualization. This file does not influence the numerical processing or the resulting georeferencing.

4. Output products and formats

All output products are written to *Cases/Case/outputs* and are organized to support both intermediate inspection and subsequent use by the DfW algorithm.

- **__Vid_resampled.npy**: A cached NumPy array containing the grayscale video sequence after temporal resampling to the prescribed frame rate. This file is used to avoid repeated decoding and resampling of the raw video data.
- **__Initial-selection.npy**: A NumPy file storing the pixel coordinates of the ground control points as manually selected in the first video frame. These coordinates serve as the reference initialization for subsequent GCP tracking.
- **__gcps_px_predefined.npy**: A NumPy array containing the temporally tracked GCP pixel coordinates for all frames, with dimensions $T \times N_{gcp} \times 2$, where T denotes the number of frames and N_{gcp} the number of ground control points.
- **Visualization.mp4**: A diagnostic video illustrating the georeferenced frames overlaid on the provided basemap. This product is intended solely for qualitative assessment of the georeferencing performance.
- **Processed.mp4**: The final co-registered grayscale video, in which all frames have been resampled onto a single global world-coordinate grid. This standardized output constitutes the primary video input for the DfW depth-inversion algorithm.
- **extent.txt**: A text file specifying the global world-coordinate extent of *Processed.mp4*, formatted as $(x_{\min}, x_{\max}, y_{\min}, y_{\max})$. This file provides the spatial referencing information required by the DfW processing stage.

5. Detailed methods with mathematical formulation

5.1 Temporal resampling and grayscale conversion

Let the raw video be recorded at a native frame rate f_{native} (Hz). The pipeline enforces a target temporal sampling defined by a user-specified frame rate FPS. When $\text{FPS} < f_{\text{native}}$, the target sampling interval is defined as $\Delta t = 1/\text{FPS}$, otherwise the native sampling interval $\Delta t = 1/f_{\text{native}}$ is retained to avoid artificial temporal interpolation.

Video frames are sampled at discrete times

$$t_k = k\Delta t, \quad k = 0, 1, 2, \dots$$

by selecting, for each t_k , the first raw frame whose acquisition time satisfies $t \geq t_k$. This procedure ensures a monotonically increasing time sequence without frame duplication.

Each selected frame is then converted to an 8-bit grayscale representation. This conversion enforces a consistent radiometric format across all frames and reduces data volume, thereby facilitating efficient and robust downstream processing in the subsequent georeferencing and warping stages.

5.2 Pitch-roll rectification by planar homography

Attitude-induced geometric distortions associated with camera pitch and roll are corrected through a planar homography that maps image pixels from the raw image plane to a rectified image plane. Let a pixel in the original image be represented in homogeneous coordinates as

$$\mathbf{p} = [u, v, 1]^\top$$

and let the corresponding rectified pixel location be denoted by $\mathbf{p}_r = [u_r, v_r, 1]^\top$. The rectification is defined, up to a scale factor, by the homographic relation

$$\mathbf{p}_r \sim \mathbf{H}\mathbf{p}$$

The homography matrix $\mathbf{H}$ is constructed as

$$\mathbf{H} = \mathbf{K}\mathbf{R}_{\text{cam}}\mathbf{K}^{-1}$$

where $\mathbf{K}$ is the camera intrinsic matrix and $\mathbf{R}_{\text{cam}}$ is the rotation matrix describing the camera attitude.

The intrinsic matrix $\mathbf{K}$ is defined from the provided focal length and the image center, assuming a pinhole camera model, as

$$\mathbf{K} = \begin{bmatrix} f & 0 & u_0 \\ 0 & f & v_0 \\ 0 & 0 & 1 \end{bmatrix}$$

where f is the focal length expressed in pixel units and (u_0, v_0) denotes the principal point, taken as the image center.

The rotation matrix $\mathbf{R}_{\text{cam}}$ accounts solely for pitch and roll, under a fixed body-to-camera axis convention. Yaw is intentionally excluded, such that the rectification removes only attitude-induced tilting effects while preserving the in-plane orientation of the image.

To control the output spatial resolution and limit memory usage, an isotropic scaling matrix

$$\mathbf{S} = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

is applied to the warped image, where the scale factor s is chosen based on the desired output resolution. The four corners of the warped image are first projected using $\mathbf{H}$ and $\mathbf{S}$, and a translation matrix

$$\mathbf{T} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$$

is then computed such that the entire warped image lies within the bounds of the output raster.

The final transformation applied during rectification is therefore given by

$$\mathbf{H}_{\text{warp}} = \mathbf{T} \mathbf{S} \mathbf{H}$$

This formulation ensures that pitch and roll effects are removed in a geometrically consistent manner, while providing explicit control over the output image size and resolution.

5.3 Interactive GCP selection and tracking

Ground control points are manually selected on the first rectified frame using an interactive, zoom-enabled interface to ensure accurate initialization in pixel space. These initial GCP locations provide the reference for subsequent temporal tracking.

Temporal tracking of the GCPs is performed using a pyramidal Lucas–Kanade optical-flow algorithm. For each GCP, the inter-frame displacement δx between two consecutive frames t and $t + 1$ is estimated by locally enforcing brightness constancy. The pyramidal formulation enables robust tracking under moderate inter-frame motion by progressively refining the displacement estimate from coarse to fine spatial scales.

To mitigate tracking drift and spurious matches, a displacement-based quality control is applied. At each time step, the maximum GCP displacement is compared against the running median displacement computed over recent frames. If the deviation exceeds a user-defined ratio threshold, the tracking result is deemed unreliable. In this case, the pipeline automatically suspends propagation and prompts manual re-selection of the GCPs on the current frame. This mechanism provides an explicit failure recovery path that prevents unbounded drift and ensures bounded error propagation over long sequences.

5.4 Frame-wise georeferencing from GCP correspondences

Let $\mathbf{W}_i = [X_i, Y_i]^\top$ denote the fixed world coordinates of the N_{gcp} ground control points, and let $\mathbf{P}_i(t) = [u_i(t), v_i(t)]^\top$ represent their tracked pixel coordinates at frame t . For each frame, the pipeline estimates a two-dimensional similarity transformation that

maps pixel coordinates to world coordinates, expressed as

$$\mathbf{W} = \mathbf{A}(t)\mathbf{P} + \mathbf{t}(t), \quad \mathbf{A}(t) = s(t)\mathbf{R}(t)\mathbf{B},$$

where $\mathbf{R}(t)$ is a 2×2 rotation matrix, $s(t)$ is a scalar scale factor, $\mathbf{t}(t)$ is a two-dimensional translation vector, and $\mathbf{B}$ is a fixed matrix accounting for the inversion of the pixel y -axis so that the image coordinate convention is consistent with the orientation of the world coordinate system.

The estimation is performed on centered coordinates to decouple rotation, scale, and translation. Let $\bar{\mathbf{P}}(t)$ and $\bar{\mathbf{W}}$ denote the centroids of the pixel and world GCPs, respectively, and define the centered coordinates as

$$\tilde{\mathbf{P}}_i(t) = \mathbf{P}_i(t) - \bar{\mathbf{P}}(t), \quad \tilde{\mathbf{W}}_i = \mathbf{W}_i - \bar{\mathbf{W}}.$$

The rotation matrix $\mathbf{R}(t)$ is obtained via an SVD-based Procrustes alignment by forming the cross-covariance matrix

$$\mathbf{H}(t) = \sum_{i=1}^{N_{\text{gcp}}} \tilde{\mathbf{P}}_i(t) \tilde{\mathbf{W}}_i^\top,$$

and computing its singular value decomposition $\mathbf{H} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$. The optimal rotation is then given by

$$\mathbf{R}(t) = \mathbf{V}\mathbf{U}^\top.$$

A determinant check is applied to ensure $\det(\mathbf{R}) = +1$, thereby preventing reflections.

The scale factor $s(t)$ is computed robustly as the median of the ratios between pairwise distances in world space and the corresponding distances in pixel space,

$$s(t) = \text{median}_{i < j} \left(\frac{\|\tilde{\mathbf{W}}_i - \tilde{\mathbf{W}}_j\|}{\|\tilde{\mathbf{P}}_i(t) - \tilde{\mathbf{P}}_j(t)\|} \right),$$

which reduces sensitivity to outliers in the GCP measurements. A potential sign ambiguity in the rotation–scale combination is resolved by evaluating the summed dot product between the rotated pixel offsets and the corresponding world offsets and enforcing a consistent orientation.

Finally, the translation vector is obtained as

$$\mathbf{t}(t) = \bar{\mathbf{W}} - \mathbf{A}(t)\bar{\mathbf{P}}(t).$$

Optionally, temporal smoothing may be applied to the pixel-space GCP trajectories $\mathbf{P}_i(t)$ prior to georeferencing. This smoothing is performed using a Savitzky–Golay filter, with an odd window length determined by the user-defined parameter `MOTION_SMOOTHING`, to suppress high-frequency jitter while preserving physically meaningful motion.

5.5 World warping and global co-registration

Given the estimated world mapping $\mathbf{W} = \mathbf{M}[u, v, 1]^\top$, where $\mathbf{M}$ is a 2×3 , affine transformation matrix, each frame is warped onto a world-coordinate raster with pixel spacing equal to the prescribed ground sampling distance (GSD). The output raster size is determined from the projected image corner coordinates, and the corresponding world extent is recorded as $(x_{\min}, x_{\max}, y_{\min}, y_{\max})$.

After per-frame warping, individual frames may still occupy distinct spatial extents. The co-registration step constructs a single global world extent that encloses all frames, defines a common global grid with the same ground sampling distance (GSD), and interpolates each frame onto this grid using regular-grid interpolation. The resulting time-consistent raster stack forms a spatiotemporally aligned 3D data cube and is exported as a single grayscale video, *Processed.mp4*, together with a corresponding global extent file, *extent.txt*.

6. Practical notes and recommended parameter choices

- FPS: choose based on the target wave periods and the desired temporal resolution; set by target wave periods; typically $> 8\sim 10$ samples per dominant period; do not exceed native FPS.
- RECTIFY_GSD_RATIO: smaller values reduce output resolution after rectification and decrease memory usage; too small values may degrade feature tracking robustness.
- WINDOW_SIZE and MAX_PYRAMID_LEVEL: increase when large inter-frame motion is expected, trading runtime and memory for tracking robustness.
- MAX_PIXEL_DISPLACEMENT: tune for anomaly detection; overly strict thresholds cause frequent manual re-selections, while loose thresholds increase drift risk.
- MOTION_SMOOTHING: use when GCP trajectories contain high-frequency jitter; avoid excessive smoothing if motion contains real rapid changes (for example sharp yaw events).
- GSD: set to the desired world pixel size; smaller improves spatial detail but increases output size as $1/\text{GSD}^2$

Contact:

kim.byunguk@snu.ac.kr

<https://bkim-git.github.io/>